

Amperometric Detection of Sub-ppm Formaldehyde Using Single-Walled Carbon Nanotubes and Hydroxylamines: A Referenced Chemiresistive System

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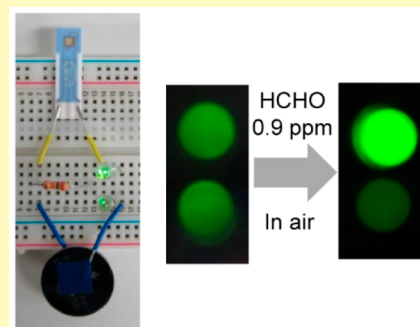
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S Supporting Information

ABSTRACT: We report amperometric detection of formaldehyde (HCHO) using hydroxylamine hydrochloride and single-walled carbon nanotubes (SWCNTs). Hydroxylamine hydrochloride reacts with HCHO to emit HCl vapor, which injects a hole carrier into semiconducting SWCNTs. The increase of conductivity in SWCNTs is easily monitored using an ohmmeter. The debundling of SWCNTs with a metallo-supramolecular polymer (MSP) increased the active surface area in the SWCNTs network, leading to excellent sensitivity to HCHO with a limit of detection (LoD) of 0.016 ppm. The response of sensor is reversible, and the sensor is reusable. The selectivity to HCHO is 10^5 – 10^6 times higher than interferences with other volatiles such as water, methanol, and toluene. Moreover, false-positive responses caused by a significant variation of humidity and/or temperature are successfully discriminated from true-positive responses by using two sensors, one with and the other without hydroxylamine hydrochloride, in a referenced system.

KEYWORDS: chemical sensors, gas sensors, chemiresistors, formaldehyde, carbon nanotubes, environmental monitoring



Formaldehyde (HCHO) is a common indoor air pollutant that can cause dermatitis, asthma, cancer, etc.¹ The World Health Organization (WHO) determines a standard of 0.08 ppm as a maximum permissible indoor HCHO concentration.² To detect HCHO, various chemical sensors³ have been developed relying on color changes,^{4,5} fluorescence,⁶ polarography,⁷ piezoelectricity,⁸ electrochemistry,⁹ and chromatography,¹⁰ for example. Although these sensors display excellent sensitivity and selectivity to HCHO (e.g., isolation of HCHO from other aldehydes and ketones), they often require bulky instruments and/or manual operation with each sampling. In this regard, conventional HCHO sensors are “detect to treat”.¹¹ However, “detect to warn” chemical sensors are in demand in a variety of areas for improving public safety, security, and health.¹²

Chemiresistive materials,¹³ displaying changes in electric resistance in response to chemical stimuli, are suited for “detect to warn” sensors because of their many advantages, including low cost, portability, real-time monitoring, and direct interface with Internet-connected electronic devices (e.g., smartphone¹⁴). Chemiresistive HCHO sensors have been realized using metal oxides.¹⁵ However, these sensors only work in temperatures over 200 °C, and thereby consume a lot of energy. Moreover, they tend to suffer from insufficient selectivity to HCHO over other volatile organic compounds.

Carbon nanotubes (CNTs), particularly single-walled carbon nanotubes (SWCNTs), are promising chemiresistive materials with operation at ambient temperatures.^{16–20} There are some examples of HCHO sensors based on functionalized CNTs.^{21–25} However, only a few examples fulfill all the requirements such as sensitivity, selectivity, and the operation at ambient atmosphere (in air, at room temperature, with humidity). Wang et al. reported that SnO₂-doped multiwalled CNTs can detect HCHO as low as 0.03 ppm under an operation temperature of 250 °C.²¹ Lu et al. utilized a CNT based sensor array to detect HCHO in air. Although the sensitivity of each sensor is limited (ca. 0.05% decrease in resistance for 0.71 ppm of HCHO), a pattern recognition algorithm based on 32 sensors is proposed to identify HCHO.²² Shi et al. reported that tetrafluorohydroquinone-mixed SWCNTs display high sensitivity to HCHO (ca. 20% increase in conductivity for 0.15 ppm of HCHO) in dry air. However, the response was almost quenched when relative humidity reached 20%.²³ Xie et al. reported that amino-group functionalized multiwalled CNTs show a 3% increase in resistance for 0.05 ppm of HCHO in pure N₂. In addition, selectivity to HCHO is ca. 10 times higher than to interferences

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with volatiles such as methanol, ammonia, and CO_2 .²⁴ As a unique approach, Kim et al. reported a field effect transistor (FET) sensor using SWCNTs covered with a thin aqueous layer.²⁵ The sensor is not negatively affected by humidity, and selectivity to HCHO is ca. 5–100 times higher than interferences with volatiles such as methanol, ammonia, and toluene.

Here we report a sensitive, selective, and reliable HCHO sensor by using SWCNTs-based chemiresistors and a HCHO-selective chemical reaction (Figure 1a). Hydroxylamine salts

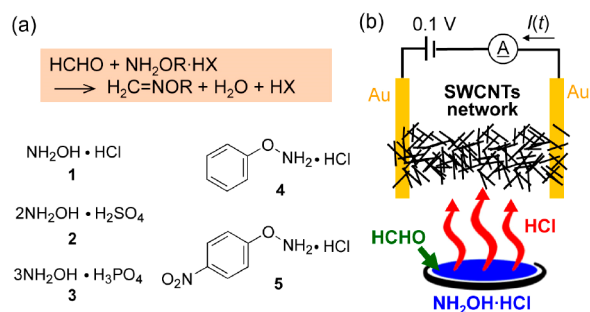


Figure 1. (a) Chemical reaction and hydroxylamine salts (1–5) used for selective detection of HCHO. (b) Illustration of sensory device constituted by SWCNTs network and 1. Electric current ($I(t)$) under 0.1 V was monitored. Gap between two Au electrodes is 200 μm .

(1–5) are employed to produce acids upon reaction with HCHO. The resulting acid is amperometrically monitored by a sensor composed of a SWCNTs network (Figure 1b). In order to realize the best sensitivity to HCHO, we optimized types of hydroxylamine salts and their support materials as well as the dispersity of SWCNTs in a network. Thus, our sensor provides an excellent limit of detection (0.016 ppm), reversibility in response, reusability, and 10^5 – 10^6 times higher selectivity to HCHO over interferences. Furthermore, false-positive responses caused by a significant variation of humidity and temperature are readily discriminated from true-positive response by using a referenced system, based on comparing signals from two sensors, with and without hydroxylamine salt.

We explored methods to combine hydroxylamine salts and SWCNTs. Commercially available gas detector tubes for HCHO were prepared by mixing hydroxylamine salts and a pH indicator to visualize sub-ppm of HCHO.²⁶ Following the traditional technique, hydroxylamine hydrochloride 1 and SWCNTs were directly mixed by drop-casting a solution of 1 (in methanol) onto a SWCNTs network. Semiconducting SWCNTs (S-SWCNTs, 95% semiconducting, HiPco), separated from as-synthesized bulk material (i.e., mixture of metallic and semiconducting SWCNTs),²⁷ were utilized since the conductivity of semiconducting SWCNTs is sensitive to electron-deficient substances, including Brønsted acids.²⁸ As shown in Figure 2a, SWCNTs drop-casted with 1 displayed some response to 3.0 ppm of HCHO in air. However, the response was unstable and irreversible. In this work, sensing responses are defined as the normalized change in electric current, described by $(I(t) - I_0)/I_0 \times 100\%$, where I_0 is the baseline electric current and $I(t)$ is the electric current at time t . This normalized procedure takes into account small differences in the initial resistance resulting from our manual fabrication process and allows for clear device-to-device comparisons.

In our next attempt, the S-SWCNTs network was covered by paper embedded with 1 (Figure S2). The sensor (S-SWCNTs/

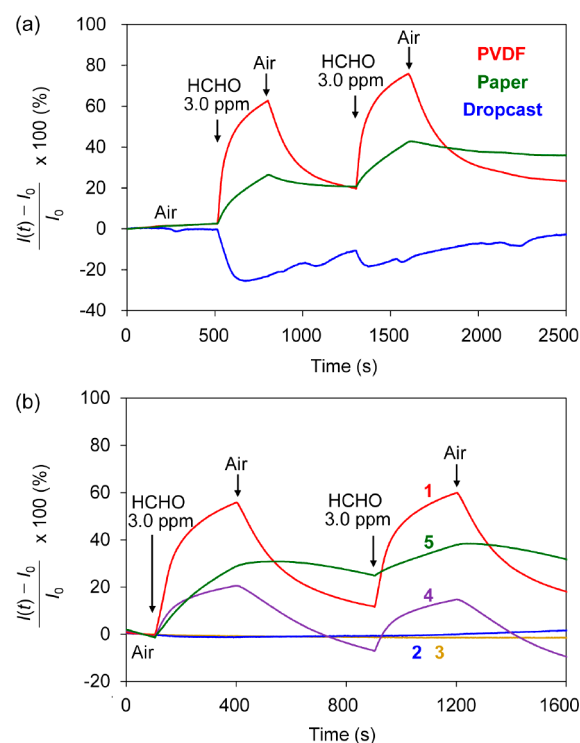


Figure 2. (a) Normalized electric current traces of S-SWCNTs sensors combined with 1 in various ways upon exposure to 3.0 ppm of HCHO in air flow (flow rate = 300 mL/min, temperature = 23 ± 1 °C, relative humidity (RH) = $38 \pm 2\%$). 1 was embedded on PVDF membrane filter, paper, or directly dropcasted on S-SWCNTs sensor. Initial resistance of each sensor is 24–40 k Ω . (b) Normalized electric current traces of S-SWCNTs sensors combined with 1–5 upon exposure to 3.0 ppm of HCHO in air flow. 1–5 was embedded on PVDF membrane filter.

1/paper) showed a much improved response to 3.0 ppm of HCHO but reversibility was not at an acceptable level. Finally, it was found that the sensor modified with 1 on a polyvinylidene difluoride membrane filter (S-SWCNTs/1/PVDF) demonstrated improved sensitivity and reversibility in response, presumably due to the chemical stability and/or hydrophobic property of PVDF. For comparison, an HCHO sensor was similarly prepared by using metallic-SWCNTs (M-SWCNTs, 90% metallic) and 1 on PVDF (Figure S5). The response was ca. 5 times smaller than that of S-SWCNTs, indicating that the semiconducting properties of SWCNTs play a main role in the sensing response.

To obtain a better selector for HCHO various types of hydroxylamine salts 2–5, listed in Figure 1a, were compared with 1 (Figure 2b). Hydroxylamine sulfate 2 and hydroxylamine phosphate 3 embedded on PVDF did not provide any response to HCHO because sulfonic acid and phosphoric acid are nonvolatile under the test conditions. Other HCl salts such as 4 and 5 worked for HCHO detection, but 1 showed better sensitivity and reversibility in response.

Further improvements in sensitivity to HCHO (ca. 2-fold) was obtained when S-SWCNTs were debundled using a metallo-supramolecular polymer (MSP, Figure S1),^{29,30} resulting in an increased active surface area of the SWCNTs network (Figures S6, S16). As shown in Figure 3a, the best sensor (S-SWCNTs/MSP/1/PVDF) displayed a highly sensitive and reversible response to HCHO down to 0.05 ppm, the lowest concentration we can reliably deliver in our gas generation

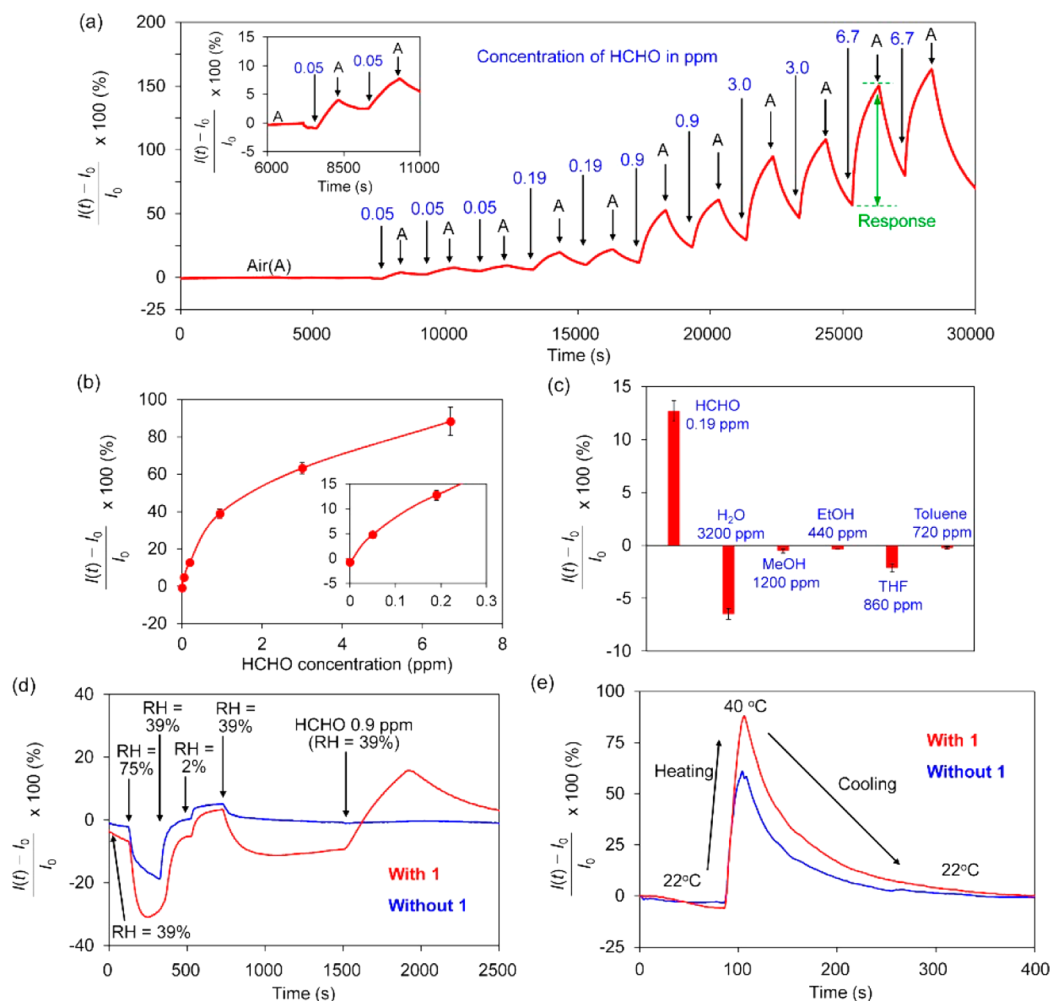


Figure 3. (a) Normalized electric current trace of S-SWCNTs/MSP/1/PVDF sensor upon exposure to HCHO in air flow (flow rate = 300 mL/min, temperature = 21.5 ± 0.8 °C, relative humidity (RH) = $36.7 \pm 1.0\%$). Concentration of HCHO ranges from 0.05 to 6.7 ppm. Exposure time and recovery time were set to 1000 s. (b) Response of S-SWCNTs/MSP/1/PVDF sensor upon exposure to HCHO at various concentration. Fitted calibration curve is shown in Figure S18. Error bars represent the standard deviation. Blank value (HCHO = 0 ppm) was obtained by using pure water instead of formalin solution. (c) Response of SWCNTs/MSP/1/PVDF sensor upon exposure to various chemical vapors in air (flow rate = 300 mL/min, temperature = 21.5 ± 0.8 °C, RH = $36.3 \pm 1.4\%$). Exposure and recovery time for HCHO is 1000 s. Exposure and recovery time for other chemicals is 100 s since responses were saturated within several tens of seconds. Error bars represent the standard deviation. (d) Normalized electric current traces of S-SWCNTs/MSP/PVDF sensors with and without 1 under significant variation of RH at 21.6 ± 0.5 °C (in air, flow rate = 300 mL/min). (e) Normalized electric current traces of S-SWCNTs/MSP/PVDF sensors with and without 1 under significant variation of temperature (in air, flow rate = 300 mL/min, RH = 39%).

system (Figure S4). The sensing response fully recovers to its original state after being left in clean air for 10 h (Figure S7a). In addition, the sensor is reusable even if the sensor is exposed to 6.7 ppm of HCHO over 12 h, indicating that our sensor has a long lifetime and the amount of 1 is in large excess compared with ppm-level HCHO vapors (Figure S7b). The correlation between the sensing response and concentration of HCHO shows a saturation curve (Figure 3b). The theoretical limit of detection (LoD), with 95% probability, is estimated to be 0.016 ppm (Figure S17).³¹

Excellent selectivity to HCHO was observed over various volatile organic compounds (Figure 3c; Figures S3, S9–S14). The magnitude of response to 0.19 ppm of HCHO is larger than that to hundred-thousands ppm of interferences with other volatiles such as water, methanol, and toluene vapors, corresponding to 10^5 – 10^6 times higher selectivity to HCHO. In addition, these interferences provided a decrease in conductivity (i.e., opposite to HCHO) since these chemicals

can inject electrons and/or swell the SWCNTs network. Acetone shows an increase in conductivity, due to a reaction with 1 (Figure S15). Thus, the discrimination of HCHO from acetone (and other aldehydes and ketones) needs to be supported by other techniques. Nevertheless, our chemiresistive sensor, which continuously monitors HCHO, is quite useful to provide a warning at an early stage of exposure. It is likely that sensitivity to HCHO is reduced by one magnitude of order under dry conditions (relative humidity (RH) = 1.5%) (Figure S8). However, sensitivity is not greatly influenced when within common range of humidity (RH = 12.5–68%).

Despite excellent selectivity to HCHO, a significant variation of humidity and/or temperature could be interpreted as a response to HCHO. For example, air often contains a significant percentage of water vapor. A variation of 10% relative humidity at 25 °C corresponds to ± 3100 ppm of water, usually 10^3 – 10^5 times larger than analytes of interest. A common solution is to utilize (i) array sensors with different

sensitivity and selectivity^{22,32} and/or (ii) external calibration using other temperature and humidity sensors. In the former case, data analysis to isolate the true signal of a target analyte is not so straightforward at practical conditions, wherein numerous parameters vary (e.g., varieties and concentration of interferants, flow rate, humidity, and temperature).³³ In the latter case, a sensing device and calibration system will be more complex. We, on the other hand, prepared a simple referenced system that utilizes two sensors, one with hydroxylamine hydrochloride (S-SWCNTs/MSP/1/PVDF) and another without hydroxylamine hydrochloride (S-SWCNTs/MSP). By comparing signals of the two sensors, the true response from HCHO can be discriminated from false-positive responses caused by interfering stimuli such as a significant variation in relative humidity (Figure 3d) and temperature (Figure 3e).

Finally, the feasibility of our HCHO sensor for broad adoption and a direct interface with omnipresent electronic devices is exemplified by constructing a simple HCHO sensor made from a minimum of components (Figures 4, S19). Two

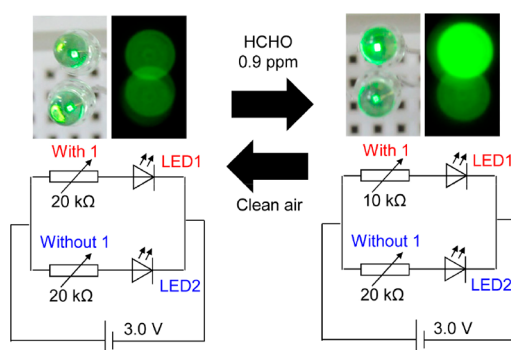


Figure 4. Simple electric device to detect formaldehyde. Photo image shows that LED1 (top) is brighter than LED2 (down) in the presence of 0.9 ppm of HCHO. Photo images taken under out-of-focus condition in the dark emphasize the differences of brightness between LED1 and LED2.

S-SWCNTs/MSP sensors, with and without **1**, were located in parallel. Each sensor is connected with a light emitting diode (LED). Initially, the resistance of the two sensors are set at approximately 20 k Ω , such that the brightness of the two LEDs look similar. When the device is exposed to 0.94 ppm of HCHO, the resistance of the S-SWCNTs/MSP with **1** decreased to ca. 10 k Ω , while the S-SWCNTs/MSP without **1** stayed nearly constant (Figure S7). The difference of 10 k Ω is sufficient to provide a visible difference in brightness between LED 1 and LED 2. Note that this referenced system can buffer a variation of humidity and temperature since the two sensors should show similar responses to these interfering effects. Although the sensitivity of the device requires further improvement to detect 0.08 ppm of HCHO (i.e., WHO standard), we expect that the device will be useful to monitor HCHO in workplaces where employees are potentially exposed to ppm-levels of HCHO vapor (formalin solution, for example, is frequently used in hospital for sterilization, cell fixing, and preservative treatment).

In conclusion, SWCNTs-based chemiresistive sensors have been successfully combined with hydroxylamine salts to produce sensitive, selective, and reliable HCHO sensors working at common atmospheric conditions. The debundling of SWCNTs by MSP improved sensitivity to HCHO with a limit of detection (LoD) reaching 0.016 ppm. The response of

sensor is reversible, and the sensor is reusable. The selectivity to HCHO is also 10^5 – 10^6 times higher than interferences with other volatiles such as water, methanol, and toluene. A simple referenced system was proposed to discriminate a true response from interfering signals. Our sensor is low cost, portable, and long lifetime, thereby making it suitable as a “detect to warn” sensory system.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssensors.7b00591.

Materials, methods, additional sensing data, data analysis, and device preparation (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Salthammer, T.; Mentese, S.; Marutzky, R. Formaldehyde in the Indoor Environment. *Chem. Rev.* **2010**, *110*, 2536–2572.
- (2) *Air Quality Guidelines for Europe*, 2nd Ed.; World Health Organization, Regional Office for Europe: Copenhagen, Denmark, 2000.
- (3) Chung, P.-R.; Tzeng, C.-T.; Ke, M.-T.; Lee, C.-Y. Formaldehyde Gas Sensors: A Review. *Sensors* **2013**, *13*, 4468–4484.
- (4) Suzuki, Y.; Nakano, N.; Suzuki, K. Portable Sick House Syndrome Gas Monitoring System Based on Novel Colorimetric Reagents for the Highly Selective and Sensitive Detection of Formaldehyde. *Environ. Sci. Technol.* **2003**, *37*, 5695–5700.
- (5) Li, Z.; Fang, M.; LaGasse, M. K.; Askim, J. R.; Suslick, K. S. Colorimetric Recognition of Aldehydes and Ketones. *Angew. Chem., Int. Ed.* **2017**, *56*, 9860–9863.
- (6) Song, H.; Rajendiran, S.; Kim, N.; Jeong, S. K.; Koo, E.; Park, G.; Thangadurai, T. D.; Yoon, S. A Tailor Designed Fluorescent ‘Turn-on’ Sensor of Formaldehyde Based on the BODIPY Motif. *Tetrahedron Lett.* **2012**, *53*, 4913–4916.
- (7) Zhang, Z. Q.; Zhang, H.; He, G. F. Preconcentration with Membrane Cell and Adsorptive Polarographic Determination of Formaldehyde in Air. *Talanta* **2002**, *57*, 317–322.
- (8) Seo, H.; Jung, S.; Jeon, S. Detection of Formaldehyde Vapor using Mercapto-phenol-Coated Piezoresistive Cantilevers. *Sens. Actuators, B* **2007**, *126*, 522–526.
- (9) Hämmerle, M.; Achmann, S.; Moos, R. Amperometric Enzyme-Based Biosensor for Direct Detection of Formaldehyde in the Gas Phase: Dependence on Electrolyte Composition. *Electroanalysis* **2008**, *20*, 410–417.

- (10) Bagheri, H.; Ghambarian, M.; Salemi, A.; Es-Haghi, A. Trace Determination of Free Formaldehyde in DTP and DT Vaccines and Diphtheria–Tetanus Antigen by Single Drop Microextraction and Gas Chromatography–Mass Spectrometry. *J. Pharm. Biomed. Anal.* **2009**, *50*, 287–292.
- (11) Grotte, J. H. Frequently Asked Questions Regarding Biological Detection; *Institute for Defense Analyses, IDA Document D-2663*, 2001.
- (12) Sferopoulos, R. A Review of Chemical Warfare Agent (CWA) Detector Technologies and Commercial-Off-The-Shelf Items; Human Protection and Performance Division, DSTO Defence Science and Technology Organisation, Australia, DSTO-GD-0570, 2009.
- (13) Neri, G. First Fifty Years of Chemoresistive Gas Sensors. *Chemosensors* **2015**, *3*, 1–20.
- (14) Azzarelli, J. M.; Mirica, K. A.; Ravnsbæk, J. B.; Swager, T. M. Wireless Gas Detection with a Smartphone via RF Communication. *Proc. Natl. Acad. Sci. U. S. A.* **2014**, *111*, 18162–18166.
- (15) Chen, D.; Yuan, Y. J. Thin-Film Sensors for Detection of Formaldehyde: A Review. *IEEE Sens. J.* **2015**, *15*, 6749–6760.
- (16) Kong, J.; Franklin, N. R.; Zhou, C.; Chapline, M. G.; Peng, S.; Cho, K.; Dai, H. Nanotube Molecular Wires as Chemical Sensors. *Science* **2000**, *287*, 622–625.
- (17) Snow, E. S.; Perkins, F. K.; Robinson, J. A. Chemical Vapor Detection using Single-walled Carbon Nanotubes. *Chem. Soc. Rev.* **2006**, *35*, 790–798.
- (18) Kauffman, D. R.; Star, A. Carbon Nanotube Gas and Vapor Sensors. *Angew. Chem., Int. Ed.* **2008**, *47*, 6550–6570.
- (19) Schnorr, J. M.; Swager, T. M. Emerging Applications of Carbon Nanotubes. *Chem. Mater.* **2011**, *23*, 646–657.
- (20) Fennell, J. F., Jr.; Liu, S. F.; Azzarelli, J. M.; Weis, J. G.; Rochat, S.; Mirica, K. A.; Ravnsbæk, J. B.; Swager, T. M. Nanowire Chemical/Biological Sensors: Status and a Roadmap for the Future. *Angew. Chem., Int. Ed.* **2016**, *55*, 1266–1281.
- (21) Wang, J.; Liu, L.; Cong, S.-Y.; Qi, J.-Q.; Xu, B.-K. An Enrichment Method to Detect Low Concentration Formaldehyde. *Sens. Actuators, B* **2008**, *134*, 1010–1015.
- (22) Lu, Y.; Meyyappan, M.; Li, J. A Carbon-Nanotube-Based Sensor Array for Formaldehyde Detection. *Nanotechnology* **2011**, *22*, 055502.
- (23) Shi, D.; Wei, L.; Wang, J.; Zhao, J.; Chen, C.; Xu, D.; Geng, H.; Zhang, Y. Solid Organic Acid Tetrafluorohydroquinone Functionalized Signle-Walled Carbon Nanotube Chemiresistive Sensors for Highly Sensitive and Selective Formaldehyde Detection. *Sens. Actuators, B* **2013**, *177*, 370–375.
- (24) Xie, H.; Sheng, C.; Chen, X.; Wang, X.; Li, Z.; Zhou, J. Multi-Wall Carbon Nanotube Gas Sensors Modified with Amino-Group to Detect Low Concentration of Formaldehyde. *Sens. Actuators, B* **2012**, *168*, 34–38.
- (25) Kim, J. Y.; Lee, J.; Hong, S.; Chung, T. D. Formaldehyde Gas Sensing Chip Based on Single-walled Carbon Nanotubes and Thin Water Layer. *Chem. Commun.* **2011**, *47*, 2892–2894.
- (26) http://www.gastec.co.jp/files/user/asset/pdf/en/detector_tube/91L.pdf (Accessed on July 4th, 2017).
- (27) Yomogida, Y.; Tanaka, T.; Zhang, M.; Yudasaka, M.; Wei, X.; Kataura, H. Industrial-Scale Separation of High-purity Single-Chirality Single-wall Carbon Nanotubes for Biological Imaging. *Nat. Commun.* **2016**, *7*, 12056.
- (28) Chandra, B.; Afzali, A.; Khare, N.; El-Ashry, M. M.; Tulevski, G. S. Stable Charge-Transfer Doping of Transparent Single-Walled Carbon Nanotube Films. *Chem. Mater.* **2010**, *22*, 5179–5183.
- (29) Ishihara, S.; Azzarelli, J. M.; Krikorian, M.; Swager, T. M. Ultratrace Detection of Toxic Chemicals: Triggered Disassembly of Supramolecular Nanotube Wrappers. *J. Am. Chem. Soc.* **2016**, *138*, 8221–8227.
- (30) Ishihara, S.; O'Kelly, C. J.; Tanaka, T.; Kataura, H.; Labuta, J.; Shingaya, Y.; Nakayama, T.; Ohsawa, T.; Nakanishi, T.; Swager, T. M. Metallic vs. Semiconducting SWCNT Chemiresistors: A Case for Separated SWCNTs Wrapped by Metallo-Supramolecular Polymer. *ACS Appl. Mater. Interfaces* **2017**, *1* DOI: 10.1021/acsami.7b12992.
- (31) Armbruster, D. A.; Pry, T. Limit of Blank, Limit of Detection and Limit of Quantitation. *Clin. Biochem. Rev.* **2008**, *29*, S49–S52.
- (32) Liu, S. F.; Moh, L. C. H.; Swager, T. M. Single-Walled Carbon Nanotube–Metalloporphyrin Chemiresistive Gas Sensor Arrays for Volatile Organic Compounds. *Chem. Mater.* **2015**, *27*, 3560–3563.
- (33) Hsieh, M.-D.; Zellers, E. T. Limits of Recognition for Simple Vapor Mixtures Determined with a Microsensor Array. *Anal. Chem.* **2004**, *76*, 1885–1895.